AP CALCULUS AB

Unit 1 • Unit 1 Fix-Up • Limits, Continuity, and Differentiability • Answer key

Unit 1 Fix-Up • Limits, Continuity, and Differentiability

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Part A • Multiple choice

Question 1

NO CALCULATOR

$\lim_{x \to -4} \dfrac{x^2 - 16}{x + 4}$ is

(A) 0

(B) -8

(C) 8

(D) nonexistent

Solution

Answer: (B).

Question 2

NO CALCULATOR

$\lim_{x \to \infty} \dfrac{2x - 7}{x^2 + 1}$ is

(A) 0

(B) 2

(C) -7

(D) ∞

Solution

Answer: (A).

Question 3

NO CALCULATOR

$\lim_{x \to 3^+} \dfrac{-1}{x-3}$ is

(A) ∞
(B) -1
(C) $-\infty$
(D) 0

Solution
Answer: (C).

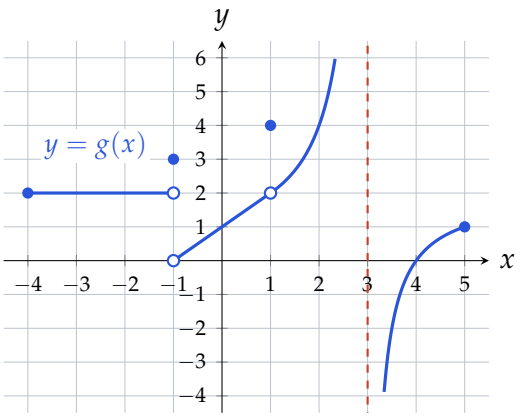
Question 4

NO CALCULATOR

$\lim_{x \to -2} \dfrac{x^2+5x+6}{x+2}$ is

(A) 0
(B) 1
(C) -1
(D) nonexistent

Solution
Answer: (B). $\frac{(x+2)(x+3)}{x+2} = x+3 \to 1$.



Question 5

NO CALCULATOR

Using the graph of g, which limit does not exist?

(A) $\lim_{x \to 1} g(x)$
(B) $\lim_{x \to 4} g(x)$
(C) $\lim_{x \to -1} g(x)$
(D) $\lim_{x \to -2} g(x)$

Solution
Answer: (C).

Question 6

NO CALCULATOR

Using the graph of g, what type of discontinuity is at $x = 3$?

(A) removable
(B) jump
(C) infinite
(D) none

Solution
Answer: (C).

Question 7

NO CALCULATOR

For which k is $f(x) = \begin{cases} 2x + k, & x < 1 \\ kx^2 + 3, & x \geq 1 \end{cases}$ continuous?

(A) 1
(B) -1
(C) It is continuous for every k.
(D) No value of k works.

Solution
Answer: (D). $2 + k = k + 3$ is impossible.

Question 8

NO CALCULATOR

Which function must have an absolute maximum on $[0, 2]$?

(A) $\frac{1}{x-1}$
(B) $\frac{1}{x+1}$
(C) $\frac{1}{x-2}$
(D) $\frac{1}{x}$

Solution
Answer: (B). Continuous on the closed interval $[0, 2]$ (EVT). The others have a vertical asymptote in $[0, 2]$.

Question 9

NO CALCULATOR

Where is $f(x) = \sqrt[3]{x - 1}$ not differentiable?

(A) $x = 1$
(B) $x = 0$
(C) $x = -1$
(D) nowhere

Solution
Answer: (A). Vertical tangent line at $x = 1$.

Question 10

NO CALCULATOR

f is differentiable with $f(2) = 4$ and $f(6) = 36$. The MVT guarantees a c in $(2, 6)$ with $f'(c) =$

(A) 3
(B) 5
(C) 8
(D) 32

Solution
Answer: (C). $\frac{36-4}{6-2} = 8$.

Part B • Free response

Question 11 (Free response)

NO CALCULATOR

Let $h(x) = \dfrac{x^2 - x - 2}{x^2 - 1}$.

(a) Find and classify each discontinuity, with limits.
(b) Find $\lim_{x \to \pm\infty} h(x)$.
(c) What value of h at the removable discontinuity makes h continuous there?

Solution
$h = \frac{(x-2)(x+1)}{(x-1)(x+1)}$. (a) $x = -1$: removable ($\lim = \frac{-3}{-2} = \frac{3}{2}$). $x = 1$: infinite ($\lim_{x \to 1^-} = +\infty$, $\lim_{x \to 1^+} = -\infty$). (b) 1. (c) $\frac{3}{2}$.

Question 12 (Free response)

NO CALCULATOR

Let $f(x) = x^3 - 4x + 1$.

(a) Use the IVT to show that f has a zero in $(0, 1)$.
(b) Find the average rate of change of f on $[0, 2]$.
(c) Explain why there is a c in $(0, 2)$ where the tangent line to the graph of f is horizontal.

Solution
(a) f is a polynomial, so it is continuous on $[0, 1]$; $f(0) = 1 > 0 > -2 = f(1)$, so the IVT gives a zero in $(0, 1)$.
(b) $\frac{f(2)-f(0)}{2-0} = \frac{1-1}{2} = 0$.
(c) f is differentiable (a polynomial), so the MVT gives c in $(0, 2)$ with $f'(c) = 0$: a horizontal tangent.